Vote Ballot

A Online Voting Platform

Ensuring Fairness, Transparency, and Efficiency in University Governance

Presentation Roadmap

Introduction

Problem Statement & Motivation

Market Analysis

Features & Functionality

System Architecture

Quality Assurance

Limitation

Conclusion

Meet Team Steadfast

The Architects Behind the Platform

Shaida Khanom Sharna

Full Stack Web Developer & Team Leader

Md Rayhanul Islam Rony

Requirement Analyst & DB Designer

Md Rayhan

Frontend Lead & UI/UX Designer

Sazzad Hossain

Associate Frontend Dev & UI Support

Israt Jahan Tamanna

QA Engineer & Backend Dev

Ismita Jahan

Maintenance Engineer & Backend Support

Project Overview

What is Vote Ballot?

Vote Ballot is a centralized web-based ecosystem engineered to automate and secure university-level elections or departmental club elections.

Core Pillars

Role-Based System

Admin, Election Authority, Student (Voter/Candidate)

System Integrity & Transparency

Ensuring tamper-resistant voting, monitoring, and verifiable

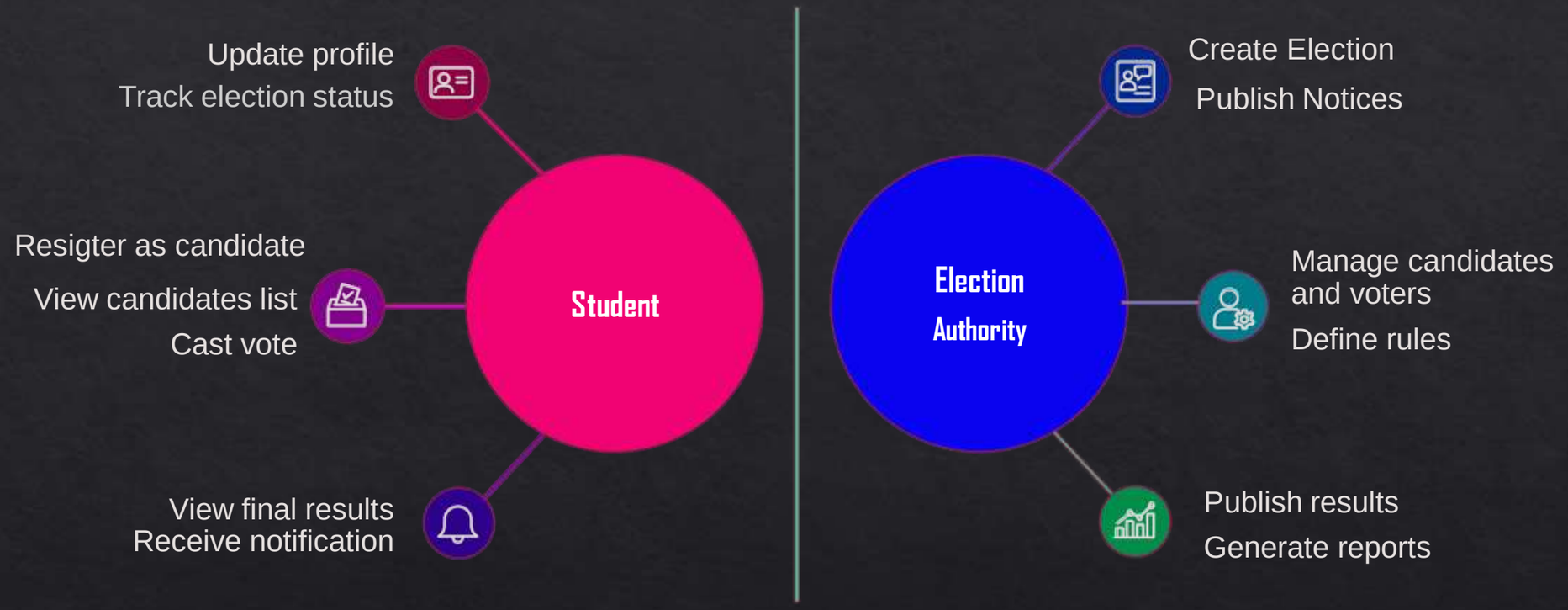
Foundational Philosophy

Security, Anonymity, and Controlled Access

Design Principle

Trust through technology-driven governance

Target Users



System Administrator (Technical Admin)

Oversee and manage all system users and roles, control account access, monitor overall system activities, maintain configurations, ensure data security and integrity, handle reported issues, and manage database backups.

Our Vision

Empowering Campus Democracy

Transform traditional manual elections into a seamless digital process

Eliminate manipulation through tamper-resistant data structures

Deliver 100% accurate, real-time election analytics

Increase student participation through accessibility and transparency

The Problem Statement

Why Change the Status Quo?

Logistical Burden

High costs for paper ballots and physical arrangements.

Rigging Risk

Vulnerable to ballot stuffing and duplicate voting.

Accessibility Issues

Off-campus students cannot participate easily.

Human Error

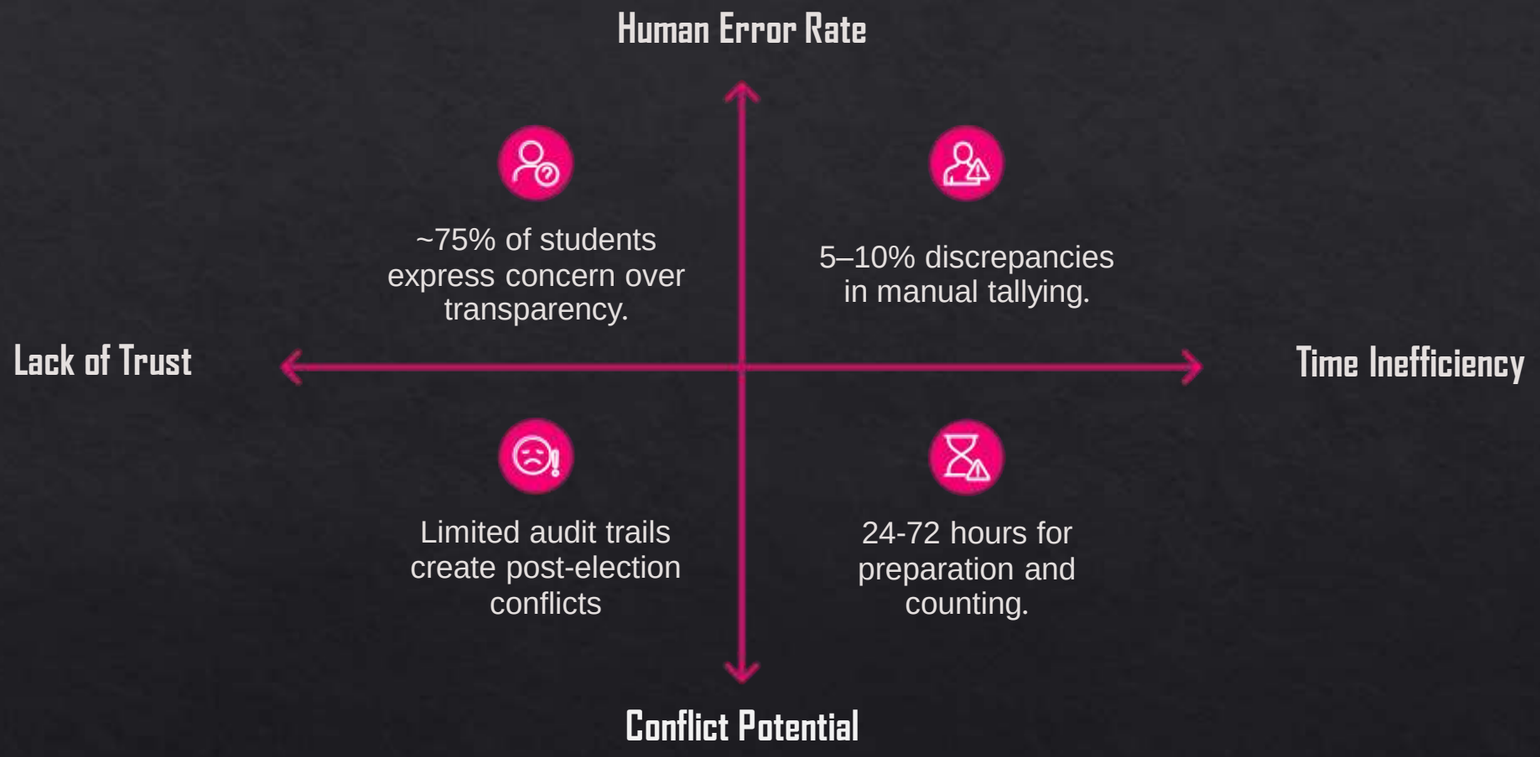
Manual counting causes mistakes and disputes.

Slow Results

Delayed announcements reduce trust.

The Problem Statement

The Integrity Gap



Our digital systems eliminate these inefficiencies instantly.

Motivation

Why Is an Online Voting Platform Necessary Today?

1

Digital Transformation

Governance systems must evolve alongside banking, commerce, and education in the digital era.

2

Remote Participation

Post-pandemic realities demand accessible and location-independent voting solutions.

3

Advanced Security Technologies

Modern encryption, secure authentication, and robust databases ensure confidentiality and ballot integrity.

The Ultimate Goal

> **Bridging trust with technology and deliver a tamper-proof, high-performance voting ecosystem that maximizes student participation by making elections accessible, secure, and instant—anytime, anywhere.**

Market Analysis & Existing Process

University Voting Landscape in Bangladesh

Current market scenario of major public universities in Bangladesh (e.g., DU, CU, RU) still rely on traditional manual voting systems for student elections.

Key Market Gaps Identified

- Significant expenditure on paper ballots and logistics
- Temporary infrastructure setup costs
- Slow result processing and delayed announcements
- Outdated, analog election workflows

Insight

While academic and financial systems have embraced digital transformation, electoral governance remains largely unchanged.

Market Analysis & Existing Process

Existing Process – Manual Election Workflow

How the Current System Operates

1. Manual verification of student IDs
2. Printing and securing thousands of ballot papers
3. Long physical queues during voting
4. Manual counting of votes
5. Result declaration after lengthy verification

Market Analysis & Existing Process

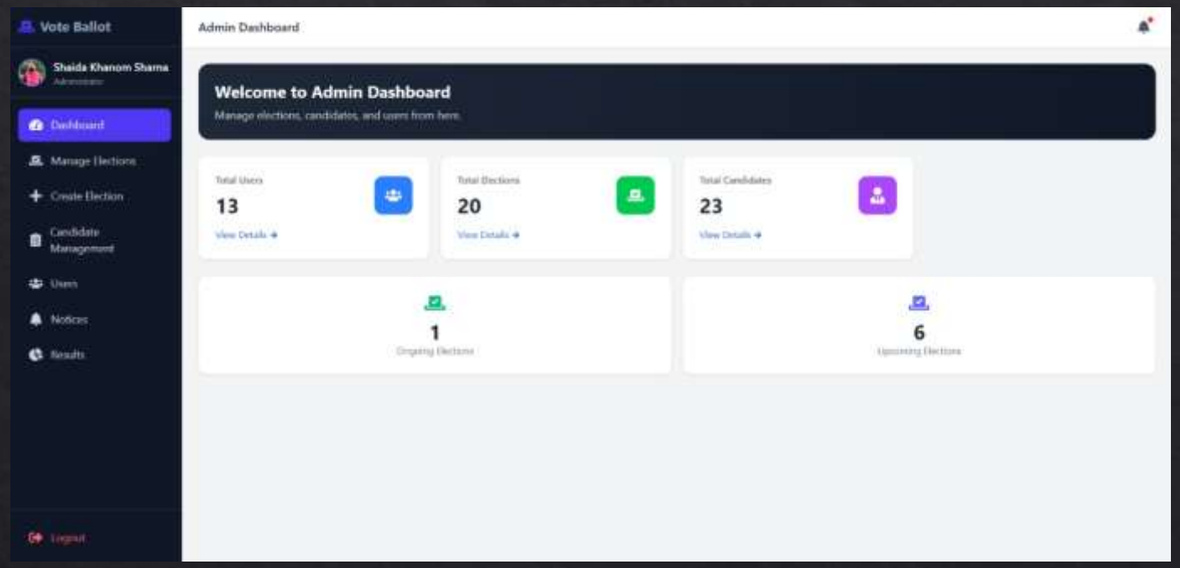
Comparative Analysis – Existing Manual System vs Vote Ballot

Aspects	Manual System	Vote Ballot (Proposed System)
Cost	High recurring expenses	Minimal cloud-based cost
Speed	24–36 hours for results	Instant automated results
Integrity	Risk of manipulation	Encrypted & tamper-resistant
Accessibility	Physical presence required	Anytime, anywhere access
Transparency	Limited audit visibility	Real-time monitoring dashboard

Features & Functionalities

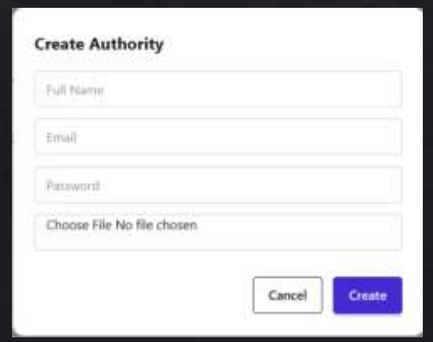
Admin Module

After secure authentication, the Admin gains centralized control over users, elections, and system-wide operations through a comprehensive management dashboard.

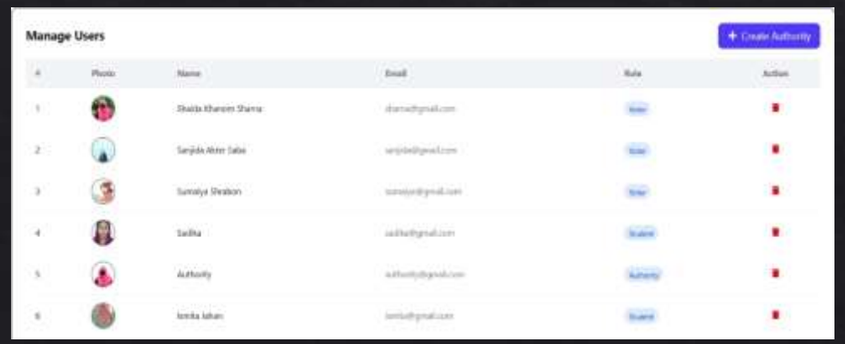


Admin Module Highlights

- **System Overview (Users, Elections, Candidates)**
- **Create Elections**
- **Manage Elections**
- **Candidate Management**
- **User Management**
- **Assign Authority**
- **Publish Notices**
- **Result Management**



Assign Authority

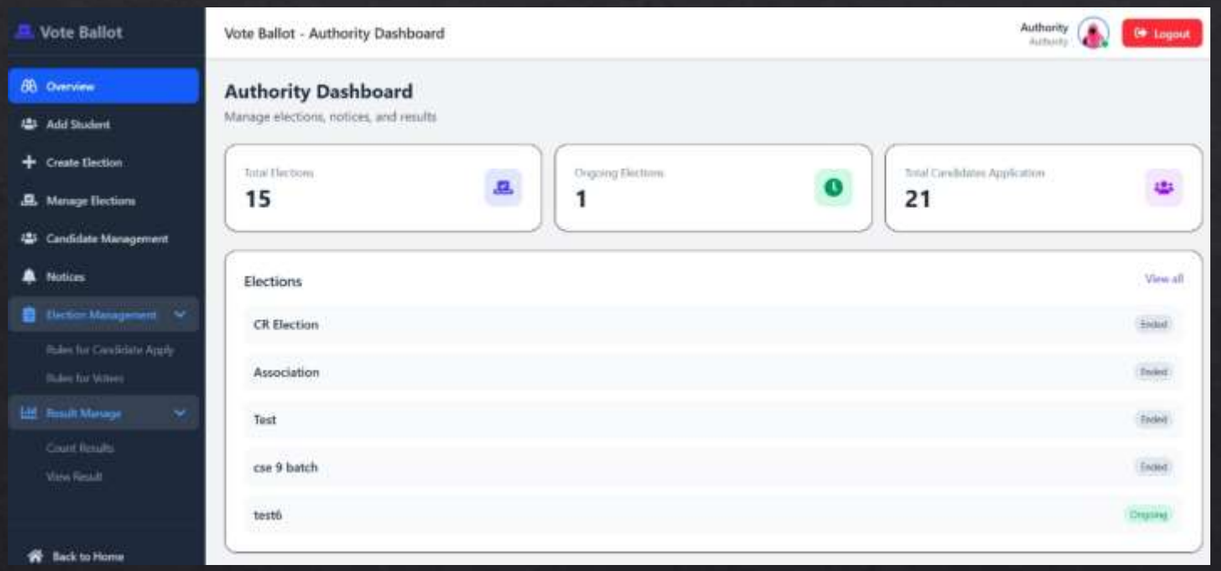


Manage all users

Features & Functionalities

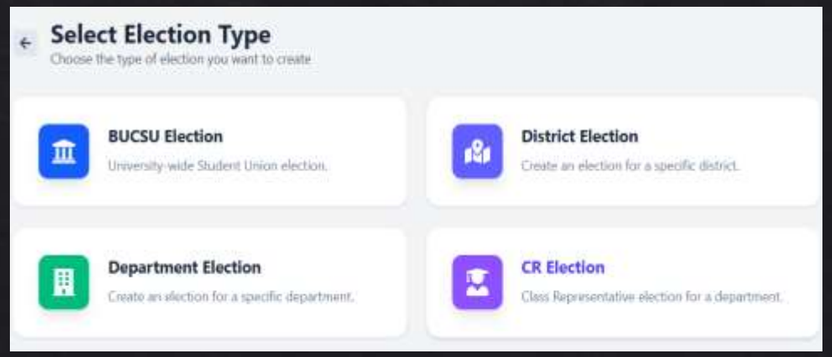
Authority Module

After secure login, authorities can manage their election activities through an interactive dashboard



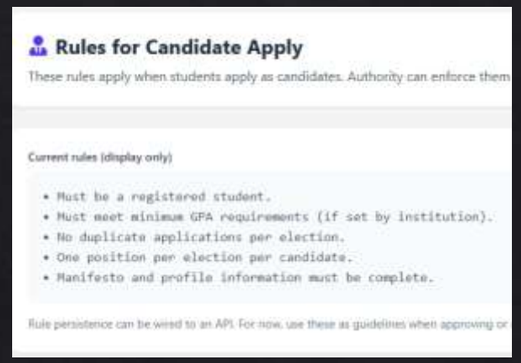
Authority Module Highlights

- **Dashboard Overview**
- **Student Management**
- **Create Elections**
- **Define Guidelines**
- **Manage Candidates & Applications**
- **Voter Management**
- **Publish Notices**
- **Result Declaration**



Select Election Types

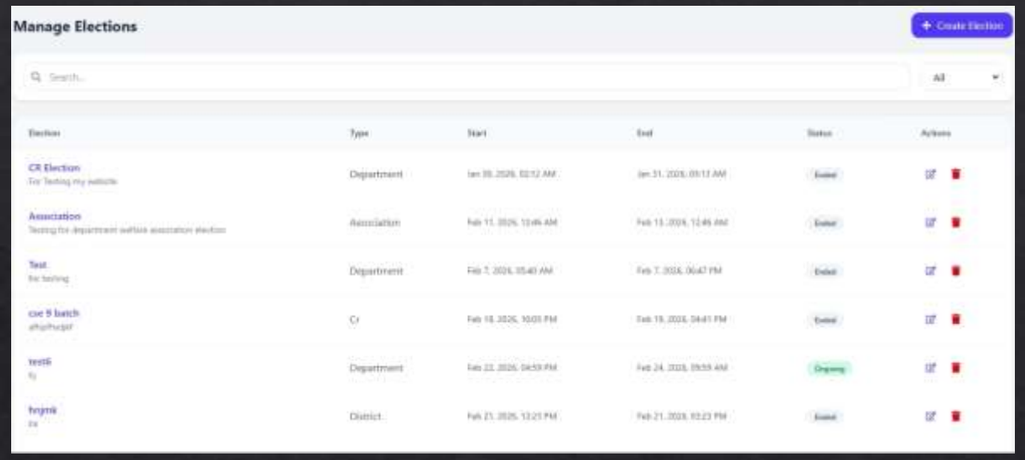
Create Election



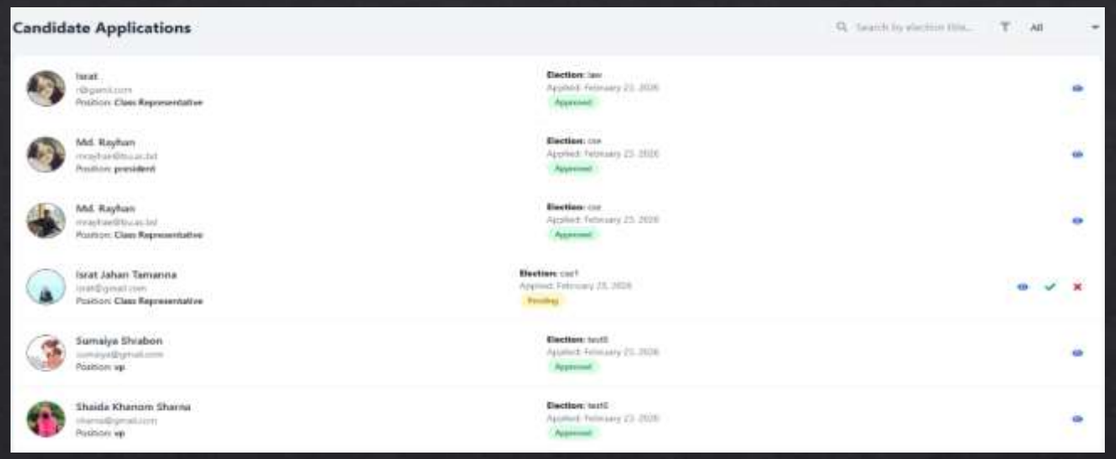
Set Guidelines

Features & Functionalities

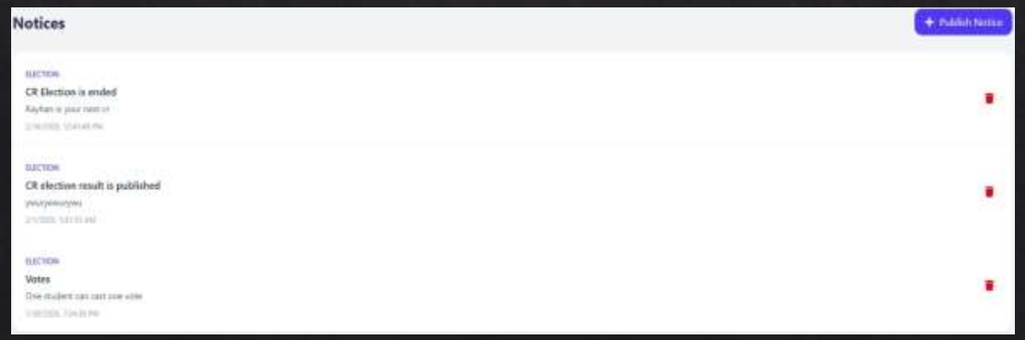
Authority Module



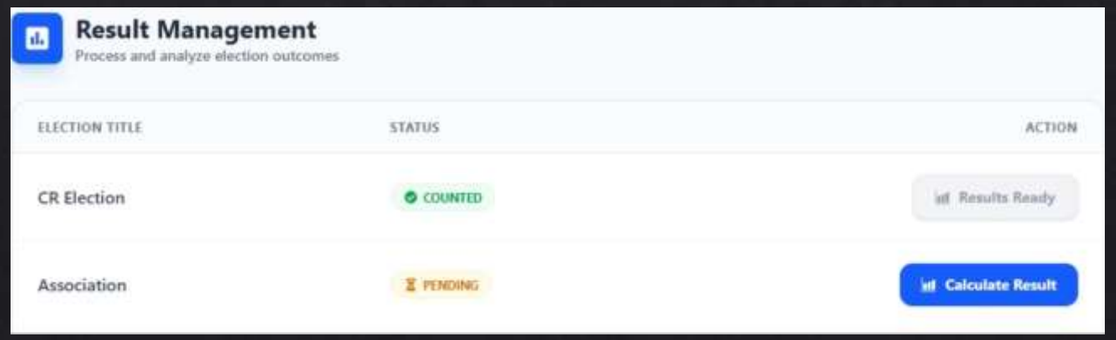
Manage Elections



Manage Candidates Application



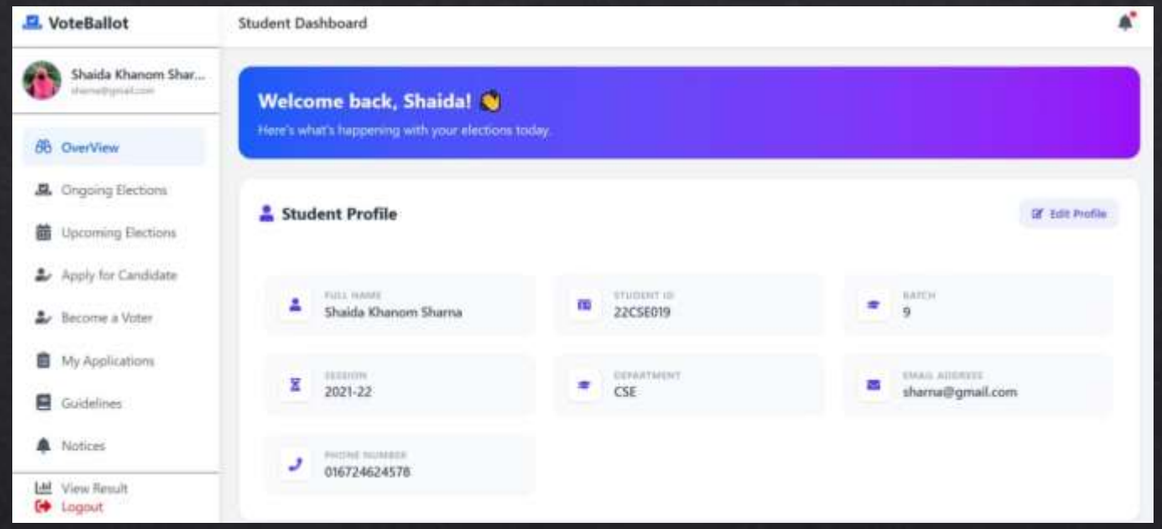
Publish Notices & Announcement



Count & Publish Results

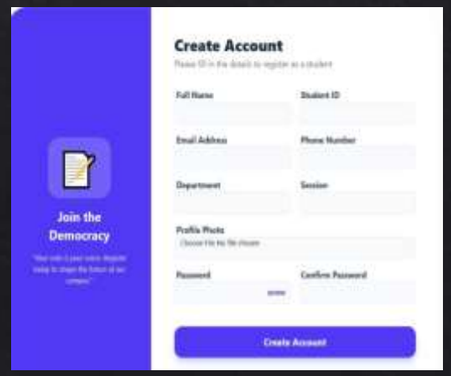
Features & Functionalities
Student Module

After secure login, students can manage their election activities through an interactive dashboard



Student Module Highlights

- **Ongoing & Upcoming Elections**
- **Apply for Candidacy**
- **Apply as Voter**
- **Application Tracking**
- **Notifications & Alerts**
- **Result Viewing**



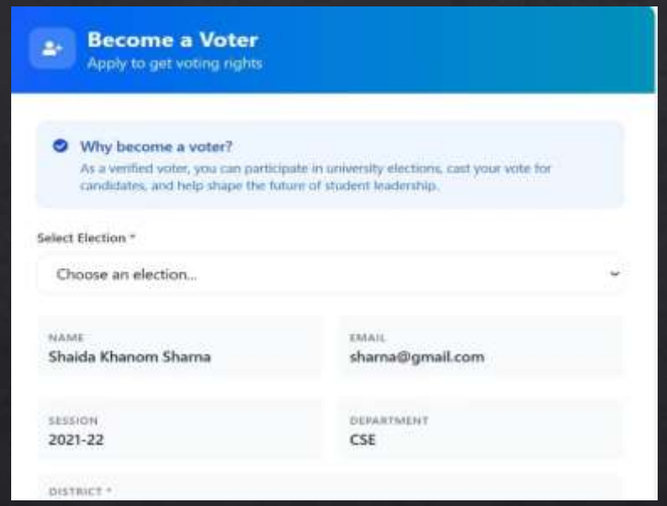
Registration & Login

See Ongoing Elections

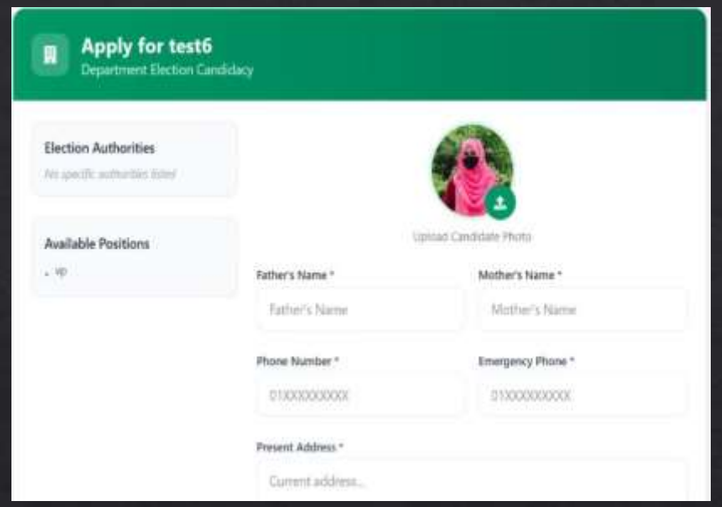
See Upcoming Elections

Features & Functionalities

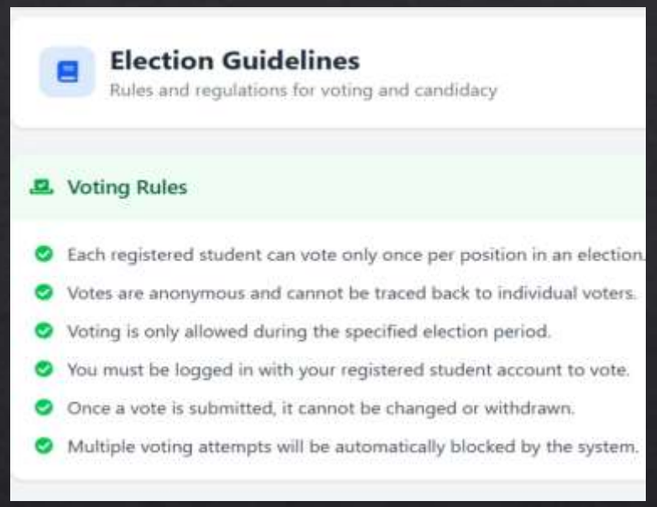
Student Module



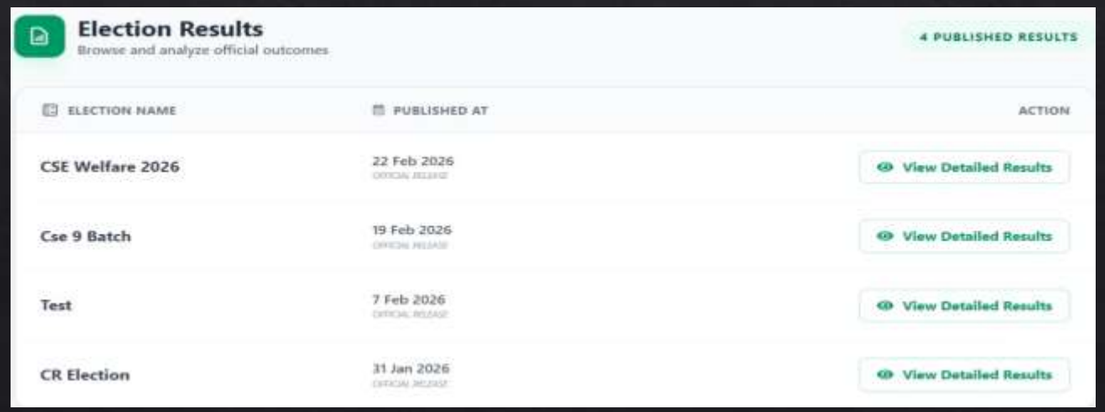
Apply for begin a voter



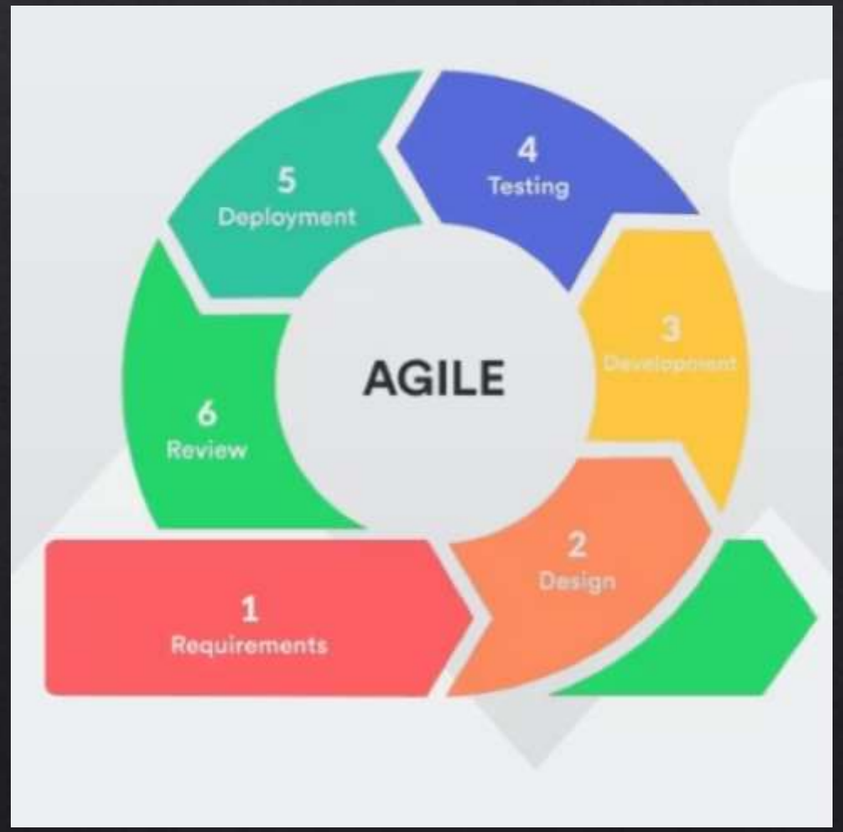
Apply for candidacy



See election rules



View elections final results



System Architecture & Design

Development Approach

Agile Methodology

The project followed an Agile approach, enabling iterative development, continuous feedback, and rapid adaptation to evolving requirements.

System Architecture & Design

Technology Stack (MERN)

Frontend
React.js & Tailwind CSS – Dynamic, component-based, responsive user interface.

Backend
Node.js & Express.js – Scalable and asynchronous request handling.

Database
MongoDB Atlas – Flexible NoSQL structure for high scalability.

Security
Crypto for secure voter info hashing.

High-Level Architecture

1. Client Layer (Browser)
2. REST API Layer
3. Business Logic (RBAC & Vote Validation
4. MongoDB Data Layer

Key Design: Strict separation of voter identity from vote data to ensure absolute anonymity.

Limitations of Vote Ballot

- Security Risks: Possibility of hacking and cyber attacks
- Technical Problems: System failure or server downtime can interrupt voting
- Digital Divide: Not all users have internet or device access

Conclusion & Future Roadmap

Vote Ballot is a secure and scalable online voting platform that enhances institutional trust by ensuring transparent, accessible, and efficient university elections.

Future Roadmap

Flutter-based mobile application (iOS & Android)

Blockchain integration for immutable vote records

Biometric Authentication Integration

“Modernizing the electoral process—one digital ballot at a time.”

Q&A

THANK YOU!

Team: Steadfast (Team No. 05)

Vote Ballot: github.com/team-steadfast/vote-ballot